The Law of Large Numbers

2026-04-17

Introduction

The law of large numbers (LLN) is the formal statement of an intuition most students bring to statistics on day one: averages settle down as you collect more data. The precise statement comes in two flavours – the weak law and the strong law – and its consequences flow through estimation theory, simulation methodology, and machine learning.

Prerequisites

The reader should know what an expectation is, what a sample mean is, and should have seen `replicate()` or a similar simulation pattern in R.

Theory

Let $X_1, X_2, \ldots$ be iid random variables with finite mean $\mu = E[X_1]$. The sample mean is $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Weak law (WLLN): for every $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

In words, the sample mean is eventually within any pre-specified distance of the population mean with probability approaching 1. This is convergence in probability: $\bar{X}_n \xrightarrow{P} \mu$.

Strong law (SLLN):

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1.$$

The sample mean converges to the population mean almost surely – for almost every realisation of the infinite sequence, the sample mean settles to μ. This is strictly stronger than the weak law.

Both laws depend on the existence of μ. For distributions without a finite mean (Cauchy, Pareto with tail index below 1), the sample mean does not converge at all; it remains wild regardless of sample size.

The LLN generalises beyond means: any continuous function of a sample mean is consistent for the same function of μ (continuous mapping theorem). This underwrites the plug-in principle and the consistency of most practical estimators.

Assumptions

The classical LLN requires iid observations with a finite mean. The LLN for non-iid sequences (stationary ergodic, martingale-difference, mixing) holds under appropriate generalisations but requires more care.

R Implementation

```r
library(ggplot2)

set.seed(2026)
mu <- 5
sigma <- 3

n_vals <- seq(10, 10000, by = 50)

draws <- rnorm(max(n_vals), mean = mu, sd = sigma)
path <- data.frame(n = 1:length(draws),
                   running_mean = cumsum(draws) / seq_along(draws))

ggplot(path, aes(x = n, y = running_mean)) +
  geom_line(colour = "steelblue") +
  geom_hline(yintercept = mu, linetype = "dashed", colour = "red") +
  scale_x_log10() +
  labs(x = "n (log scale)",
       y = expression(bar(X)[n]),
       title = "LLN in action: running mean vs. population mean") +
  theme_minimal()
```

Plotting the running sample mean against n shows it converging to $\mu = 5$; different random seeds produce different paths, all converging to the same horizontal line.

Output & Results

The plot shows an initial ragged behaviour for small n, narrowing steadily as n grows, with the running mean nested closer and closer around the true 5. Classical theoretical rates: the typical deviation at sample size n scales as $\sigma/\sqrt{n}$ (the standard error).

Interpretation

The LLN justifies every Monte Carlo estimate and every use of the sample mean as a point estimate. A paper that reports "estimated by averaging over $N = 10^5$ simulation replicates" is implicitly invoking the LLN: the average converges to the population mean and, by the CLT, has a well-characterised standard error.

Practical Tips

- The LLN guarantees convergence but says nothing about speed – the CLT does. In practice, "large enough n" depends on the population's tail behaviour.
- Check that the mean exists before trusting the LLN. A single quiet assumption failure (sampling from a heavy-tailed population) turns a routine simulation into a misleading result.
- The LLN extends to sample variances, proportions, correlations, and more general functionals – any "mean of transformed data" is consistent.
- Running-mean plots are useful diagnostics in simulations: if the running mean is still drifting at the end of the simulation, more replicates are needed.
- In online learning and streaming algorithms, the LLN underlies the convergence of stochastic-gradient and recursive estimation schemes.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr